Theorem (Riesz Representation):

$(V, \langle , \rangle)$: a inner product space

$\phi : V \longrightarrow V^*$

$\underline{v} \longmapsto \phi_{\underline{v}}$

$(\phi_{\underline{v}} : V \to \mathbb{F}, \ \phi_{\underline{v}}(\underline{w}) := \langle \underline{v}, \underline{w} \rangle$

Then:

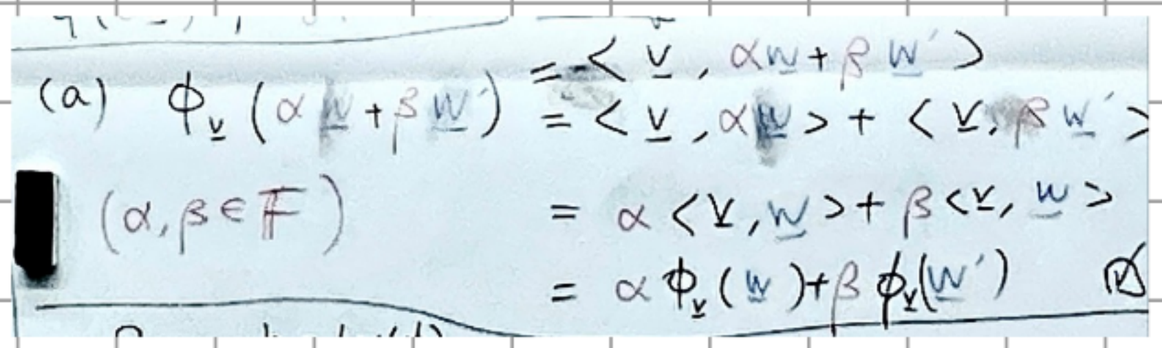
(a) $\phi_{\underline{v}} \in V^*$

(b) ϕ is injective

(c) ϕ is a $\mathbb{R}$-linear transformation
(i.e., $\forall a, b \in \mathbb{R}, \phi(a\underline{v} + b\underline{v}') = a\phi(\underline{v}) + b\phi(\underline{v}')$
and $\phi(i\underline{v}) \neq i\phi(\underline{v})$

Proof:

(a)



(b) Suppose $\underline{k} \in \ker(\phi) \Rightarrow \phi(\underline{k}) = \phi_{\underline{k}} = \underline{0}_{V^*}$

$\Rightarrow \forall \underline{w} \in V, \ \phi_{\underline{k}}(\underline{w}) = \underline{0}_{V^*}(\underline{w}) = 0$

$\Rightarrow \forall \underline{w} \in V, \ \langle \underline{k}, \underline{w} \rangle = 0$

Take $\underline{w} = \underline{k}$, then $\langle \underline{k}, \underline{k} \rangle = 0$

(positive definite) $\Leftrightarrow \underline{k} = \underline{0}$

(c) (WTS: $\phi(a\underline{v}+b\underline{v}') = a\phi(\underline{v})+b\phi(\underline{v}')$)

Take $\forall \underline{w} \in V$

LHS:

$\phi(a\underline{v}+b\underline{v}')(\underline{w})$

$= \langle a\underline{v}+b\underline{v}', \underline{w}\rangle$

$= \langle a\underline{v}, \underline{w}\rangle + \langle b\underline{v}', \underline{w}\rangle$

$= a\langle \underline{v}, \underline{w}\rangle + b\langle \underline{v}', \underline{w}\rangle$

RHS:

$=$

Corollary:

$$\begin{cases} V \text{ is a } \mathbb{R}\text{-inner product space} \\ \dim_{\mathbb{R}}(V) < \infty \end{cases} \Rightarrow \phi \text{ is an isomorphism} \quad (\text{def. 见上上页})$$

Proof:

$$\begin{cases} \dim(V^*) = \dim(V) < \infty \\ \phi: V \to V^* \text{ is inj.} \end{cases} \Rightarrow \phi \text{ is bij.}$$

关于 $\mathbb{C}$-inner product space V:

ϕ 不是 $\mathbb{C}$-linear trans.

To remedy: treat V as a $\mathbb{R}$-inner product space

e.g.: $V = \mathbb{C}^n = \mathrm{Span}_{\mathbb{C}}\{\underline{e}_1, \cdots, \underline{e}_n\}$

At the same time: treat $V = \mathbb{C}^n$ as a $\mathbb{R}$-vector space

e.g.: $\underline{v} = \begin{pmatrix} a_1 + b_1 i \\ \vdots \\ a_n + b_n i \end{pmatrix} \in \mathbb{C}^n \quad (a_i, b_i \in \mathbb{R})$

$$= a_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + b_1 \begin{pmatrix} i \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \cdots + a_n \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} + b_n \begin{pmatrix} 0 \\ \vdots \\ 0 \\ i \end{pmatrix}$$

$$\Downarrow$$

$$\underline{v} \in \mathrm{Span}_{\mathbb{R}}\{\underline{e}_1, i\underline{e}_1, \cdots, \underline{e}_n, i\underline{e}_n\}$$

In general, if $\dim_{\mathbb{C}}(V) = n$ with basis $V = \mathrm{Span}_{\mathbb{C}}\{\underline{v}_1, \cdots, \underline{v}_n\}$

Then $V_{\mathbb{R}}$ is also a $\mathbb{R}$-v. space of $\dim_{\mathbb{R}}(V_{\mathbb{R}}) = 2n$ with basis $\{\underline{v}_1, i\underline{v}_1, \cdots, \underline{v}_n, i\underline{v}_n\}$.

Similarly, a $\mathbb{C}$-inner product space $(V, \langle , \rangle_{\mathbb{C}})$ can be seen as a $\mathbb{R}$-inner product space $(V_{\mathbb{R}}, \langle , \rangle_{\mathbb{R}})$.

Then $\phi: V_{\mathbb{R}} \to (V_{\mathbb{R}})^*$ will become an isom. of $\mathbb{R}$-vector spaces

$$\wr$$

$\phi: V \to V^*$ is a bijection (but not $\mathbb{C}$-linear)

Adjoint Operator

Def.: ~~~~~~~~ (?)

Lemma:

Let $(V, \langle , \rangle)$ be a finite dimensional linear inner product space over ($\mathbb{R}$ or $\mathbb{C}$)

$T: V \to V$ be a linear operator

$\Downarrow$

Then $\exists$ unique $a_{\underline{v}} \in V$, s.t. $\langle a_{\underline{v}}, \underline{w} \rangle = \langle \underline{v}, T(\underline{w}) \rangle$, $\forall \underline{w} \in V$

Proof:

Consider $\gamma \in V^*$, $\gamma = \phi_{\underline{v}} \circ T$

i.e., $\gamma : \underset{\in V}{\underline{w}} \xmapsto{T} \underset{\in V}{T(\underline{w})} \xmapsto{\phi_{\underline{v}}} \underset{\in \mathbb{H}}{\langle \underline{v}, T(\underline{w}) \rangle}$

By Riesz Representation, there's bijection

$$\phi : V \to V^*$$

$\Rightarrow$ unique $\underline{a}_{\underline{v}} \in V$ so that

$$\underline{a}_{\underline{v}} \longmapsto \phi_{\underline{a}_{\underline{v}}} = \gamma$$

$$\therefore \phi_{\underline{a}_{\underline{v}}}(\underline{w}) = \gamma(\underline{w}) \quad \forall \underline{w} \in V$$

$$\Rightarrow \langle \underline{a}_{\underline{v}}, \underline{w} \rangle = \langle \underline{v}, T(\underline{w}) \rangle \quad \square$$

Adjoint Map

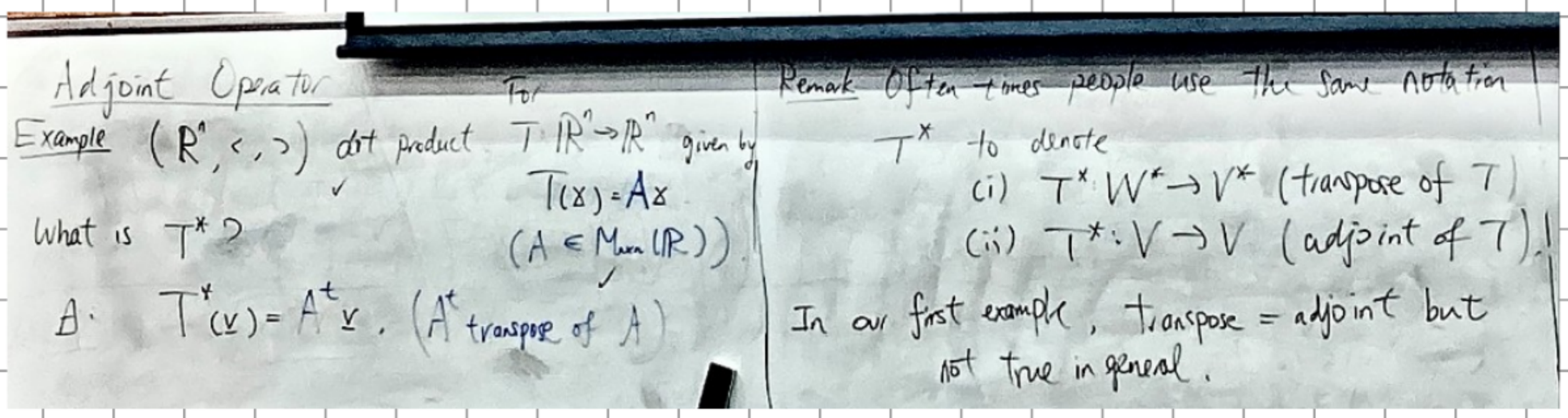
Ref.:

$(V, \langle , \rangle)$

adjoint map of T: $T^* : V \to V$, $\underline{v} \mapsto \underline{a}_{\underline{v}}$, $T^*(\underline{v}) := \underline{a}_{\underline{v}}$

i.e., $\forall \underline{v}, \underline{w} \in V$, $\langle T^*(\underline{v}), \underline{w} \rangle = \langle \underline{v}, T(\underline{w}) \rangle$

Adjoint Operator



Adjoint Operator

Example $(\mathbb{R}^n, \langle , \rangle)$ dot product ✓

What is T^*?

A: $T^*(\underline{v}) = A^t \underline{v}$. ($A^t$ transpose of A)

For $T: \mathbb{R}^n \to \mathbb{R}^n$ given by $T(x) = Ax$ ($A \in M_{n\times n}(\mathbb{R})$) ✓

Remark Often times people use the same notation T^* to denote

(i) $T^*: W^* \to V^*$ (transpose of T)

(ii) $T^*: V \to V$ (adjoint of T).

In our first example, transpose = adjoint but not true in general.

Remark:

transpose of T: $T^*: W^* \to V^*$

$\neq$

adjoint of T: $T^*: V \to V$

E.g. 2: $\mathbb{C}^n$, $\langle \underline{v}, \underline{w} \rangle = (\overline{v_1}, \cdots, \overline{v_n}) \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} = \underline{v}^t \underline{w}$

$T(\underline{x}) = A\underline{x}$, where $A \in M_{n\times n}(\mathbb{C})$

Claim: $T^*(\underline{v}) = \overline{A}^t \underline{v}$

LHS:

$$\langle T^*(\underline{v}), \underline{w} \rangle = \overline{(\overline{A}^t \underline{v})^t} \underline{w}$$

$$= \overline{\underline{v}^t (\overline{A}^t)^t} \underline{w}$$

$$= \overline{\underline{v}^t} A \underline{w}$$

RHS: $\langle \underline{v}, T(\underline{w}) \rangle = \underline{v}^t (A\underline{w}) = \underline{v}^t A \underline{w} = \text{LHS}$

Prop.: The adjoint map $T^*: V \to V$ is a linear trans.

Proof: (WTS: $T^*(\alpha \underline{v} + \beta \underline{v}') = \alpha T^*(\underline{v}) + \beta T^*(\underline{v}')$

(WTS: $T^*(\alpha \underline{v} + \beta \underline{v}') - (\alpha T^*(\underline{v}) + \beta T^*(\underline{v}')) = \underline{0}$)

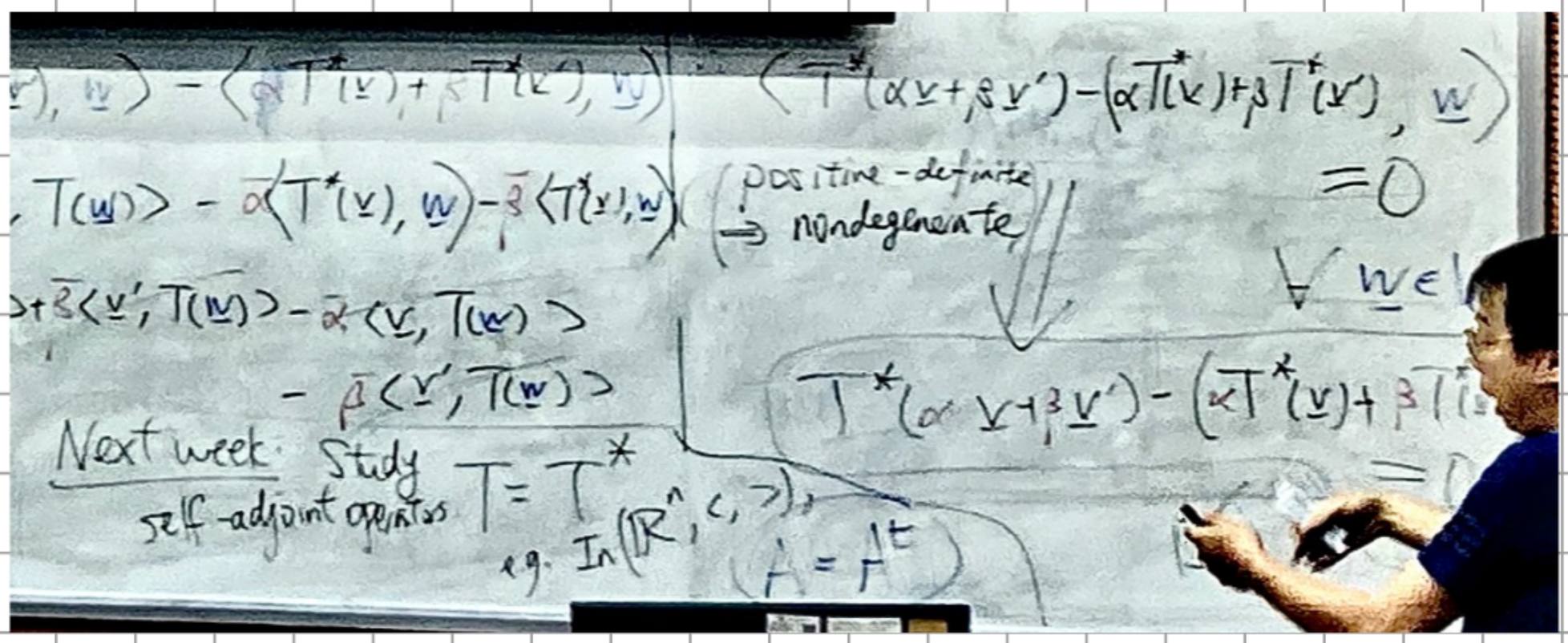
Consider $\langle T^*(\alpha \underline{v} + \beta \underline{v}') - (\alpha T^*(\underline{v}) + \beta T^*(\underline{v}')), \underline{w} \rangle$, $(\exists \underline{w} \in V)$

$= \langle \sim, \underline{w} \rangle - \langle \sim, \underline{w} \rangle$

$= \langle \alpha \underline{v} + \beta \underline{v}', T(\underline{w}) \rangle - \overline{\alpha} \langle T^*(\underline{v}), \underline{w} \rangle - \overline{\beta} \langle \sim \rangle$

$= \overline{\alpha} \langle \underline{v} + T(\underline{w}) \rangle + \sim - \sim - \sim$

$= \underline{0}$


$\langle \underline{v}', \underline{w} \rangle - \langle \alpha T^*(\underline{v}) + \beta T^*(\underline{v}'), \underline{w} \rangle$
$\langle \underline{v}, T(\underline{w}) \rangle - \bar{\alpha} \langle T^*(\underline{v}), \underline{w} \rangle - \bar{\beta} \langle T^*(\underline{v}'), \underline{w} \rangle$
$+ \bar{\beta} \langle \underline{v}', T(\underline{w}) \rangle - \bar{\alpha} \langle \underline{v}, T(\underline{w}) \rangle$
$- \beta \langle \underline{v}', T(\underline{w}) \rangle$
Next week: Study self-adjoint operators $T = T^*$
e.g. $\ln(\mathbb{R}^n, \langle , \rangle)$, $A = A^t$
$\langle T^*(\alpha \underline{v} + \beta \underline{v}') - (\alpha T^*(\underline{v}) + \beta T^*(\underline{v}'), \underline{w} \rangle$
$= 0$
(positive-definite $\Rightarrow$ nondegenerate)
$\forall \underline{w} \in$
$T^*(\alpha \underline{v} + \beta \underline{v}') - (\alpha T^*(\underline{v}) + \beta T^*$
$= 0$